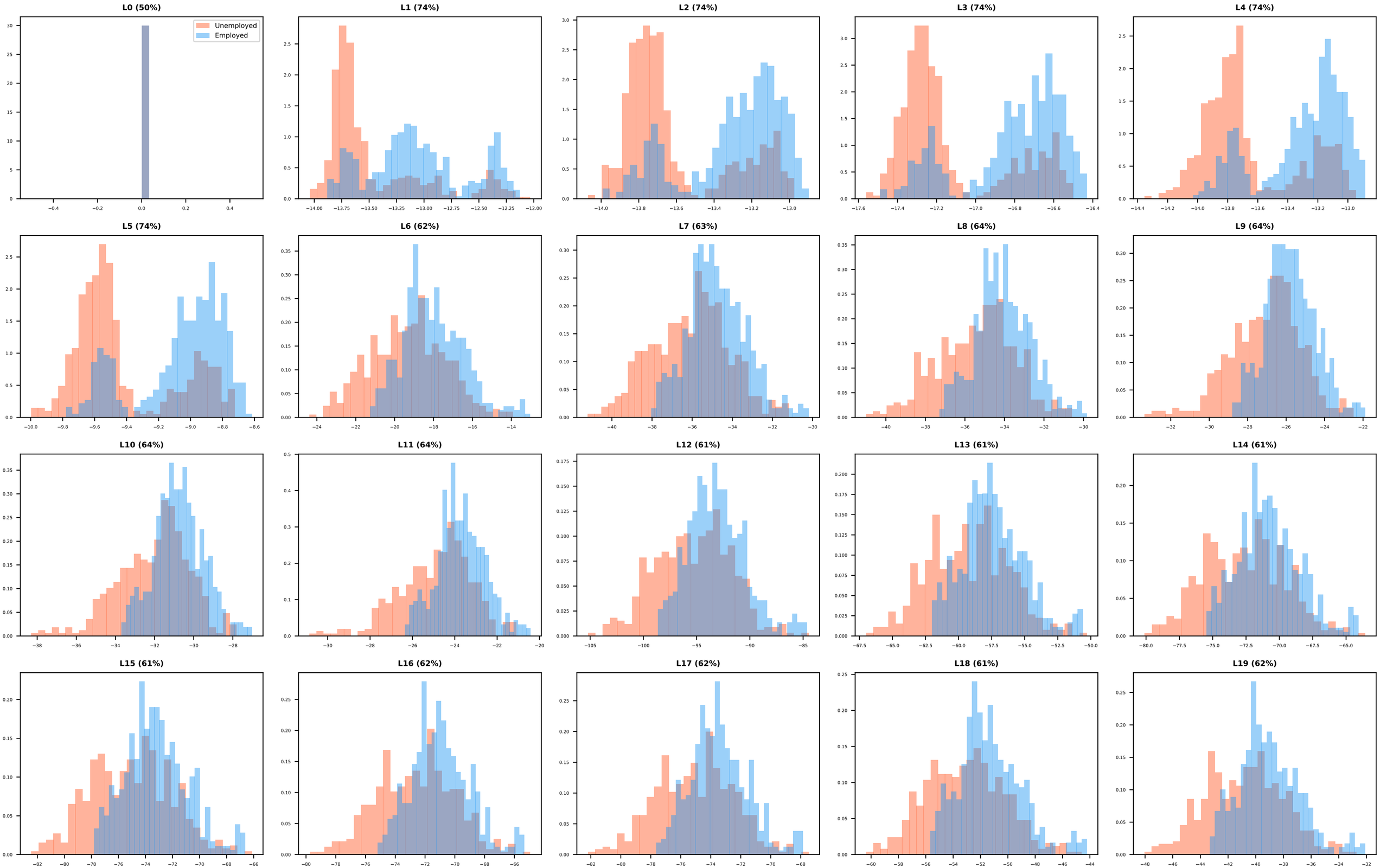
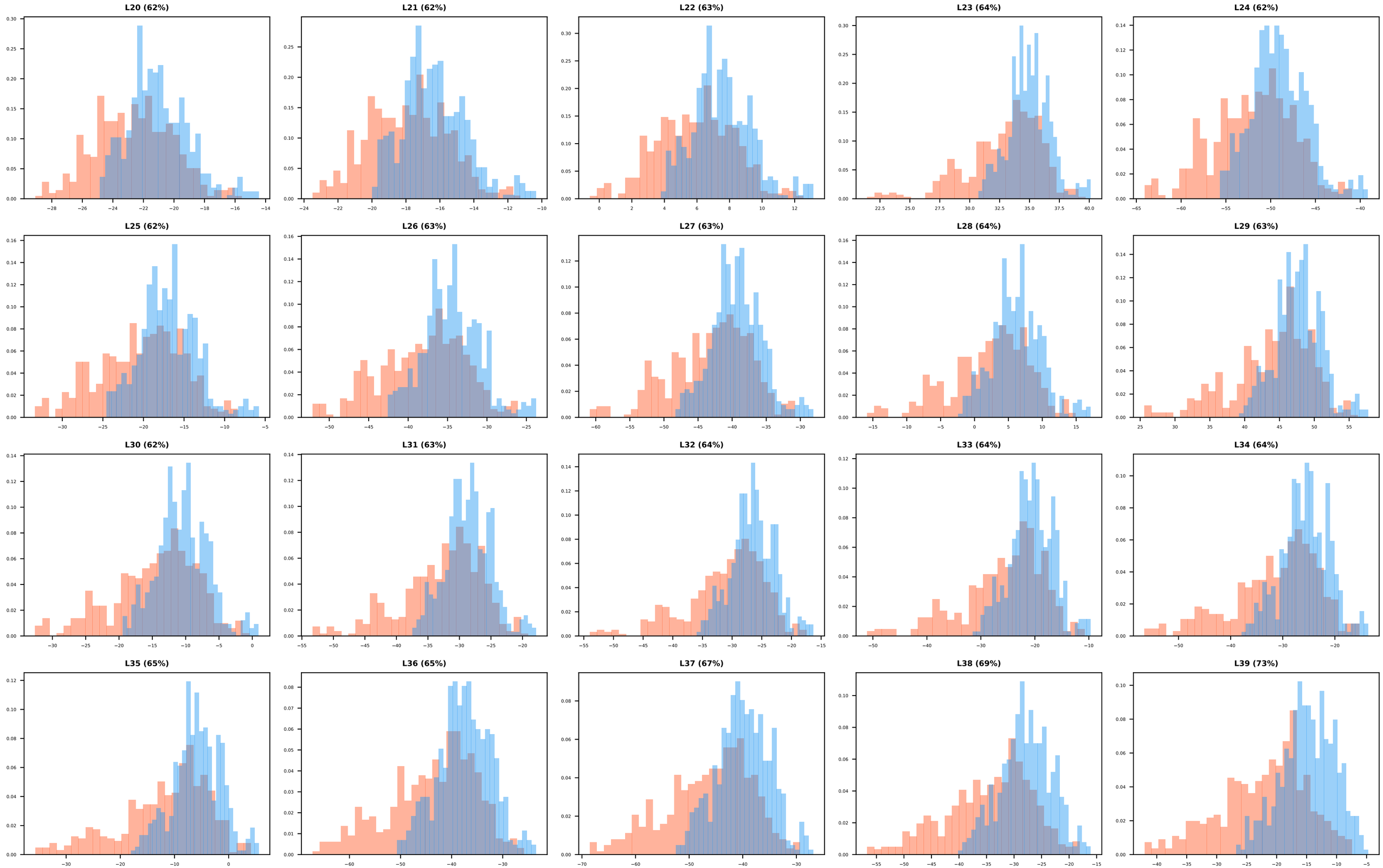
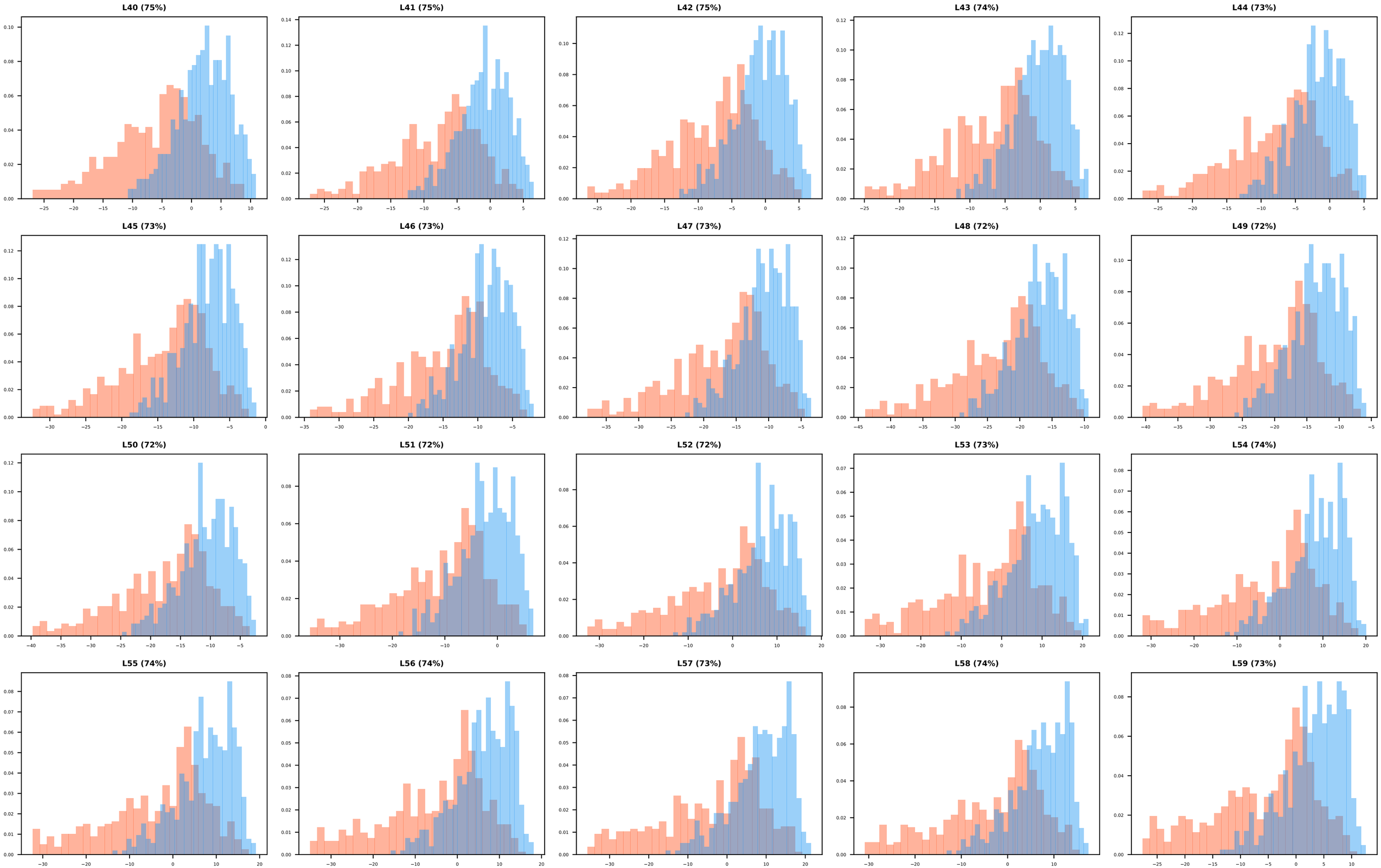
Projection Distribution — V8/exp2/cfg2 (geo=orig_day=orig_prompt=cot) — Page 1



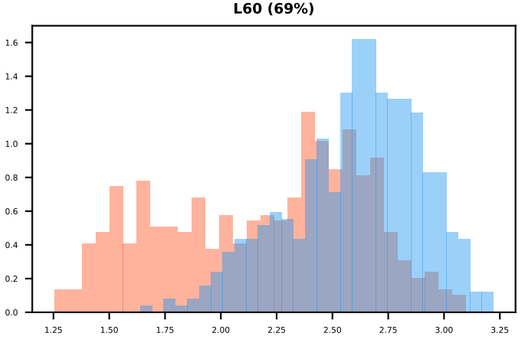
Projection Distribution — V8/exp2/cfg2 (geo=orig_day=orig_prompt=cot) — Page 2



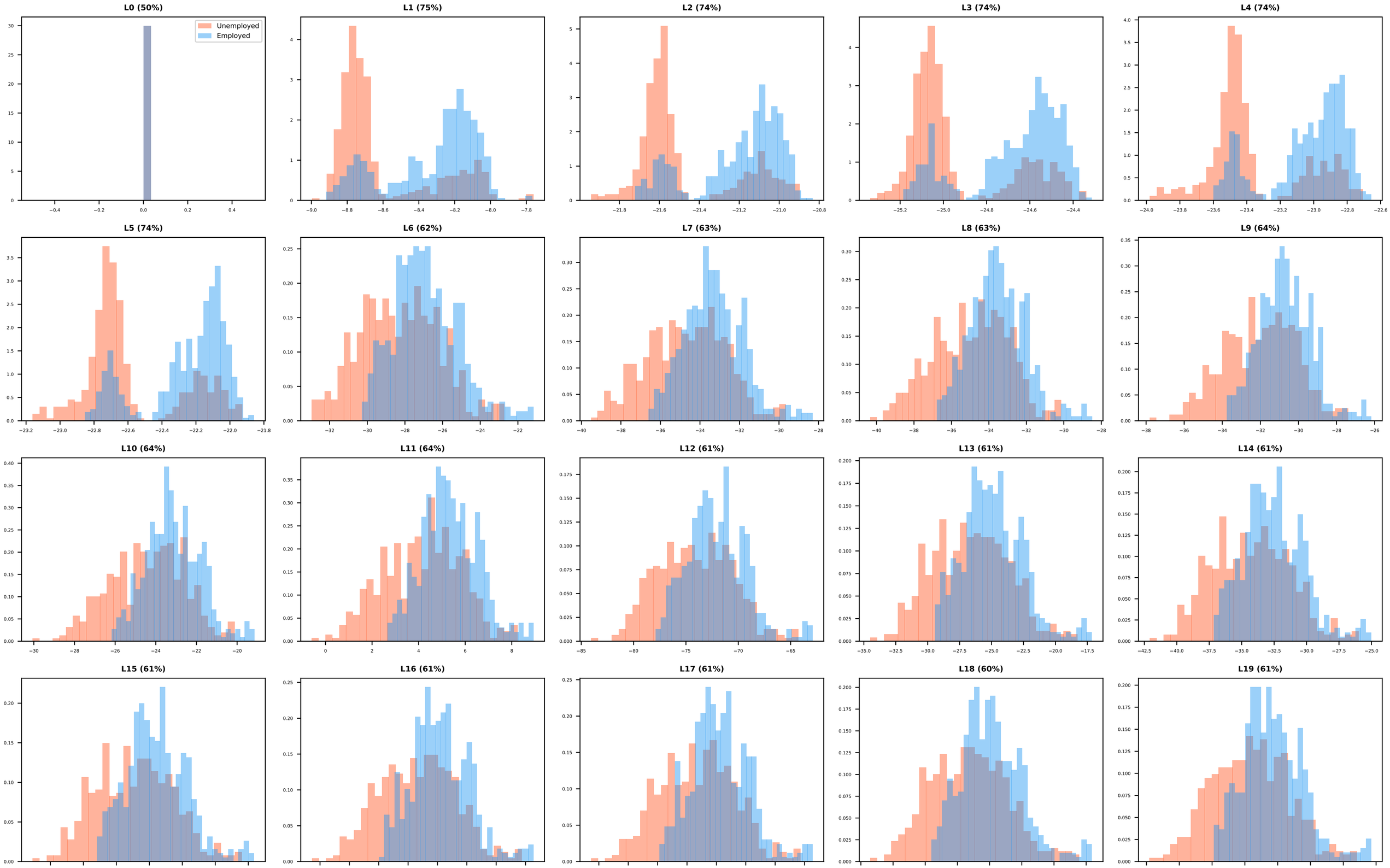
Projection Distribution — V8/exp2/cfg2 (geo=orig_day=orig_prompt=cot) — Page 3



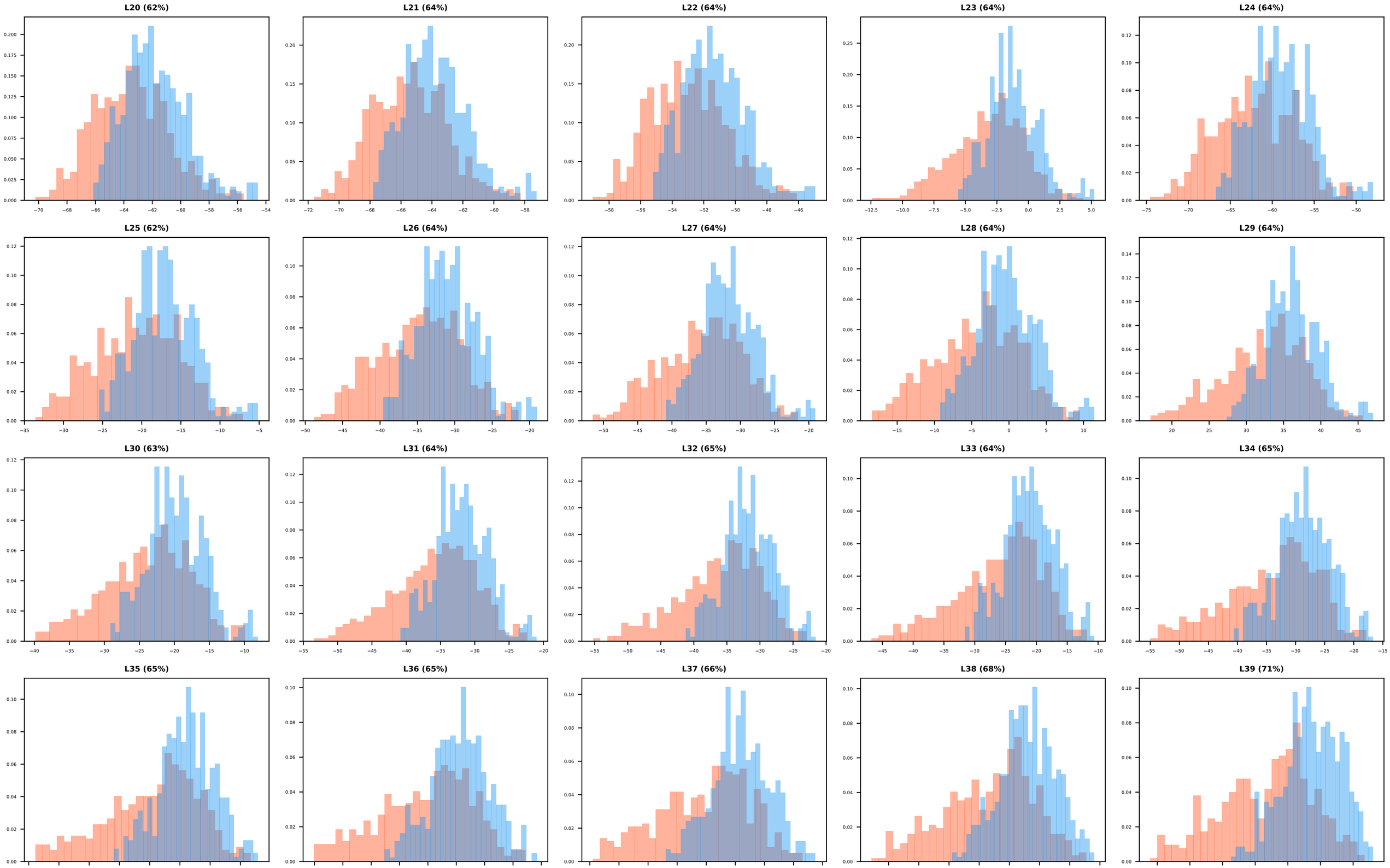
Projection Distribution — V8/exp2/cfg2 (geo=orig_day=orig_prompt=cot) — Page 4



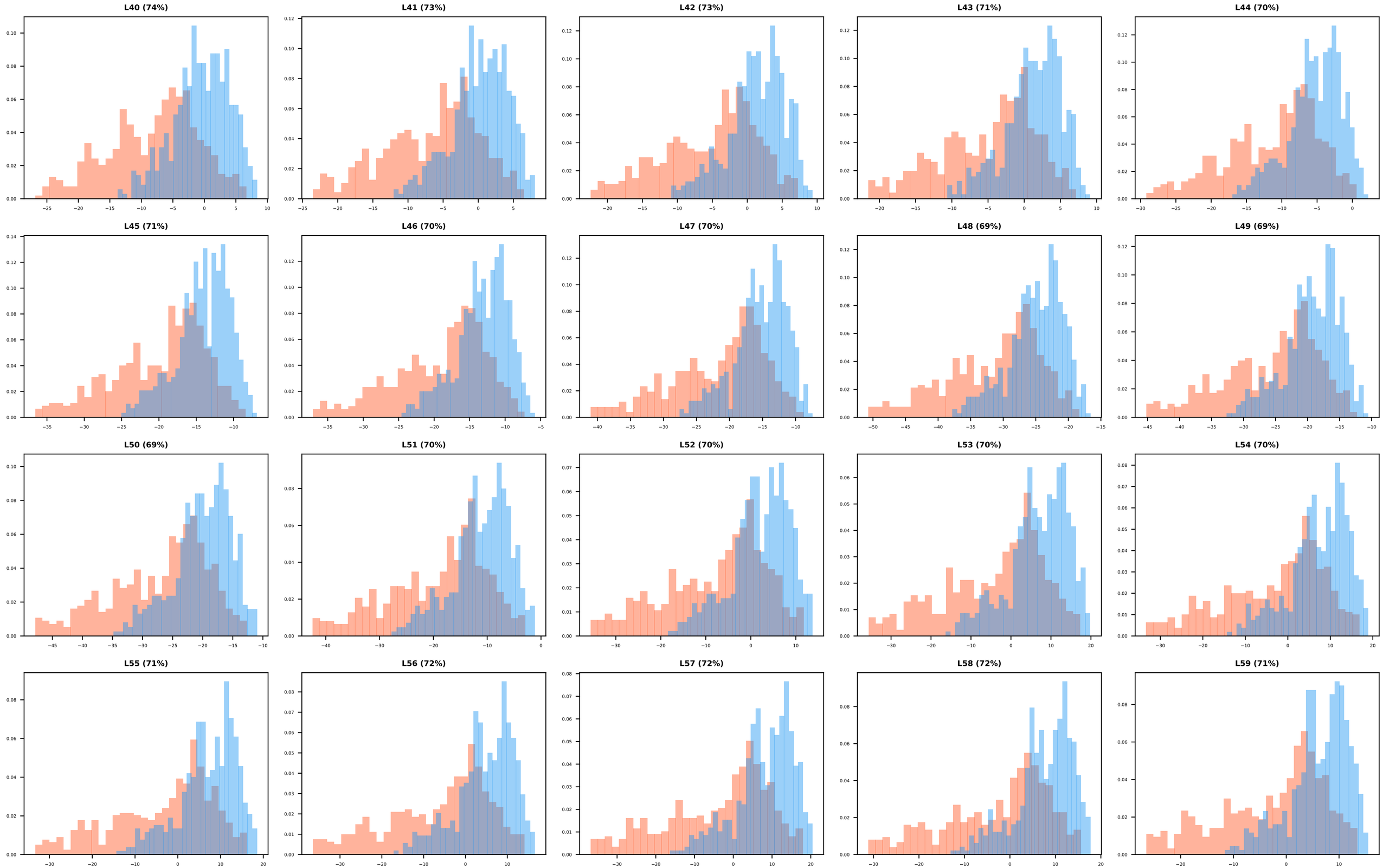
Projection Distribution — V8/exp2/cfg4 (geo=orig_day=hash_prompt=cot) — Page 1



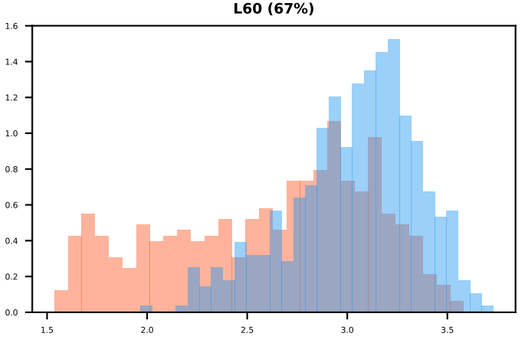
Projection Distribution — V8/exp2/cfg4 (geo=orig_day=hash_prompt=cot) — Page 2



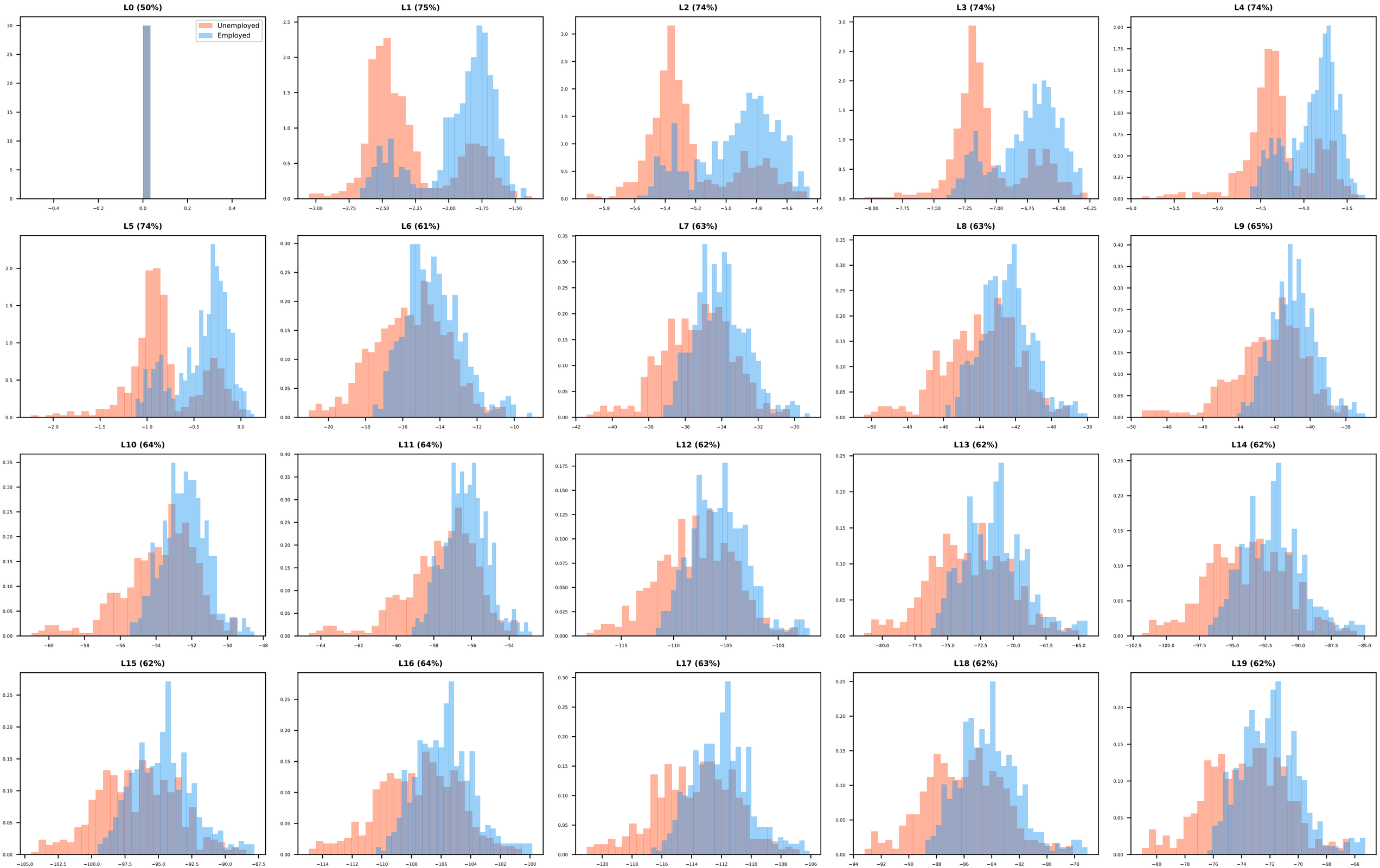
Projection Distribution — V8/exp2/cfg4 (geo=orig_day=hash_prompt=cot) — Page 3



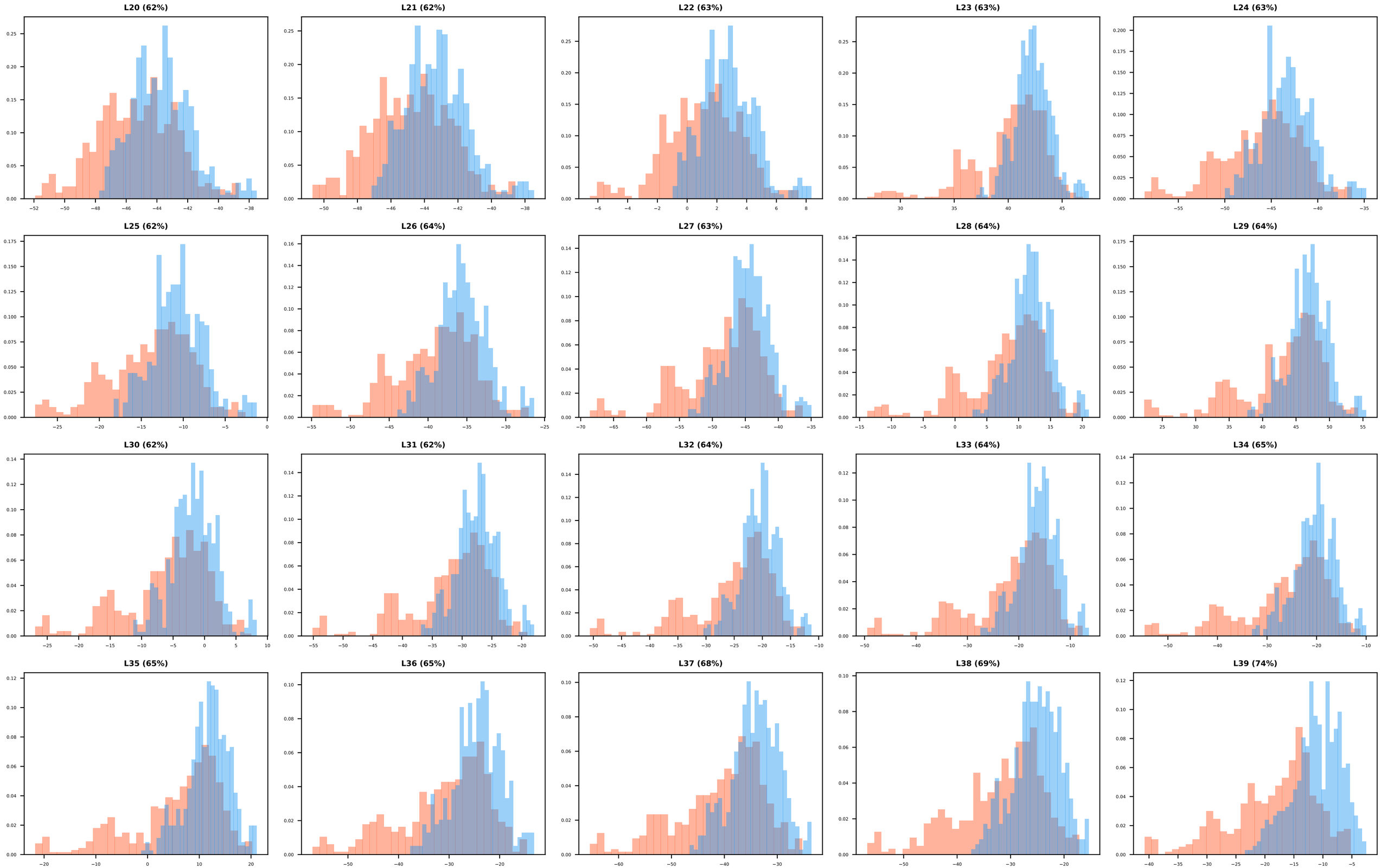
Projection Distribution — V8/exp2/cfg4 (geo=orig_day=hash_prompt=cot) — Page 4



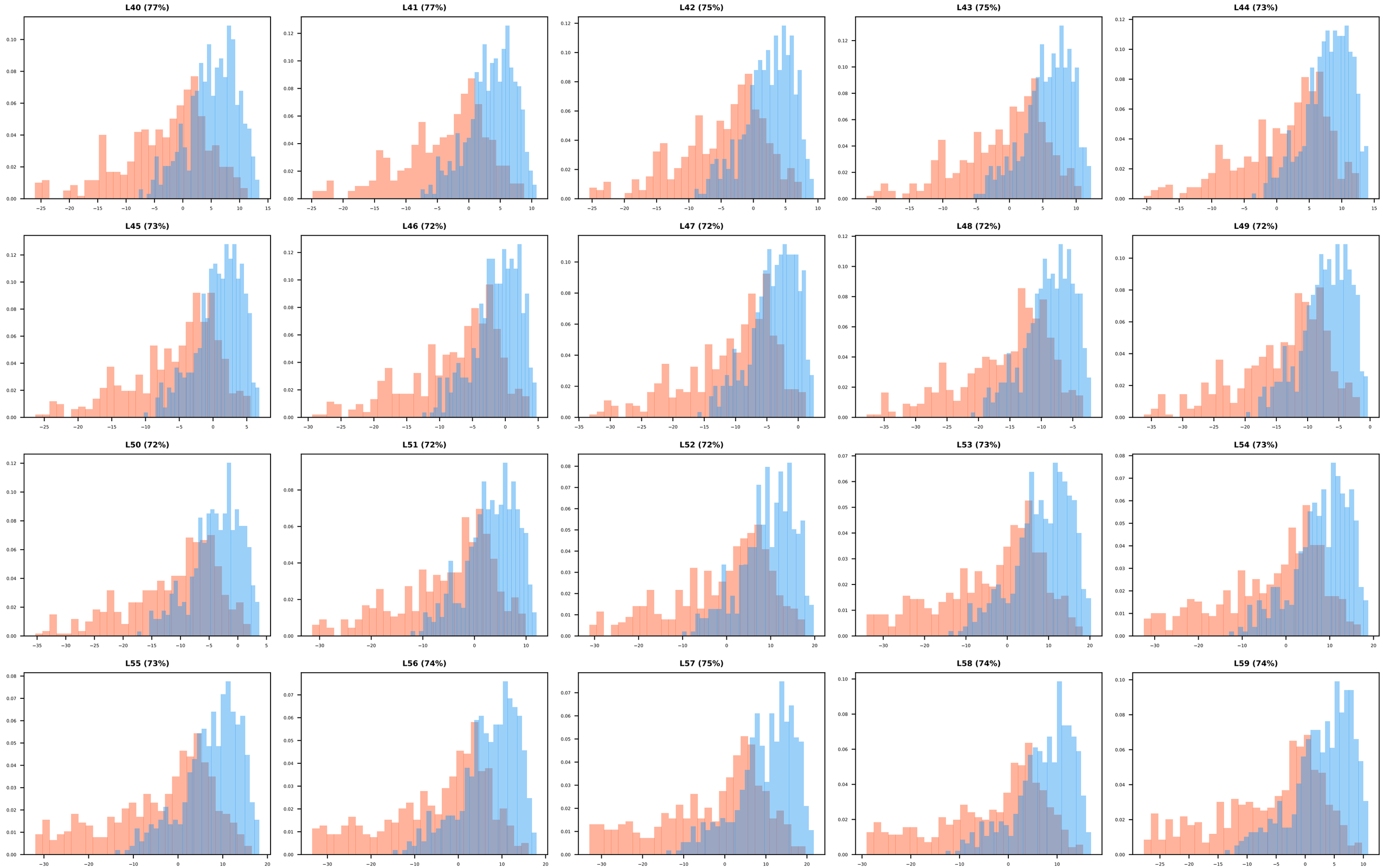
Projection Distribution — V8/exp2/cfg6 (geo=hash_day=orig_prompt=cot) — Page 1



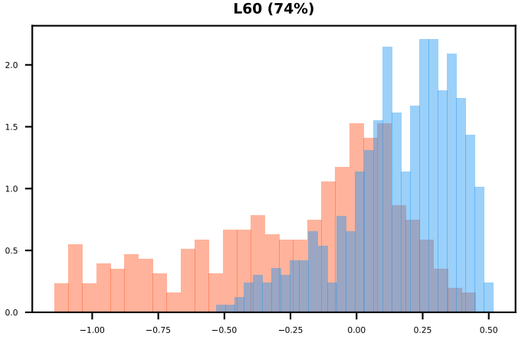
Projection Distribution — V8/exp2/cfg6 (geo=hash_day=orig_prompt=cot) — Page 2



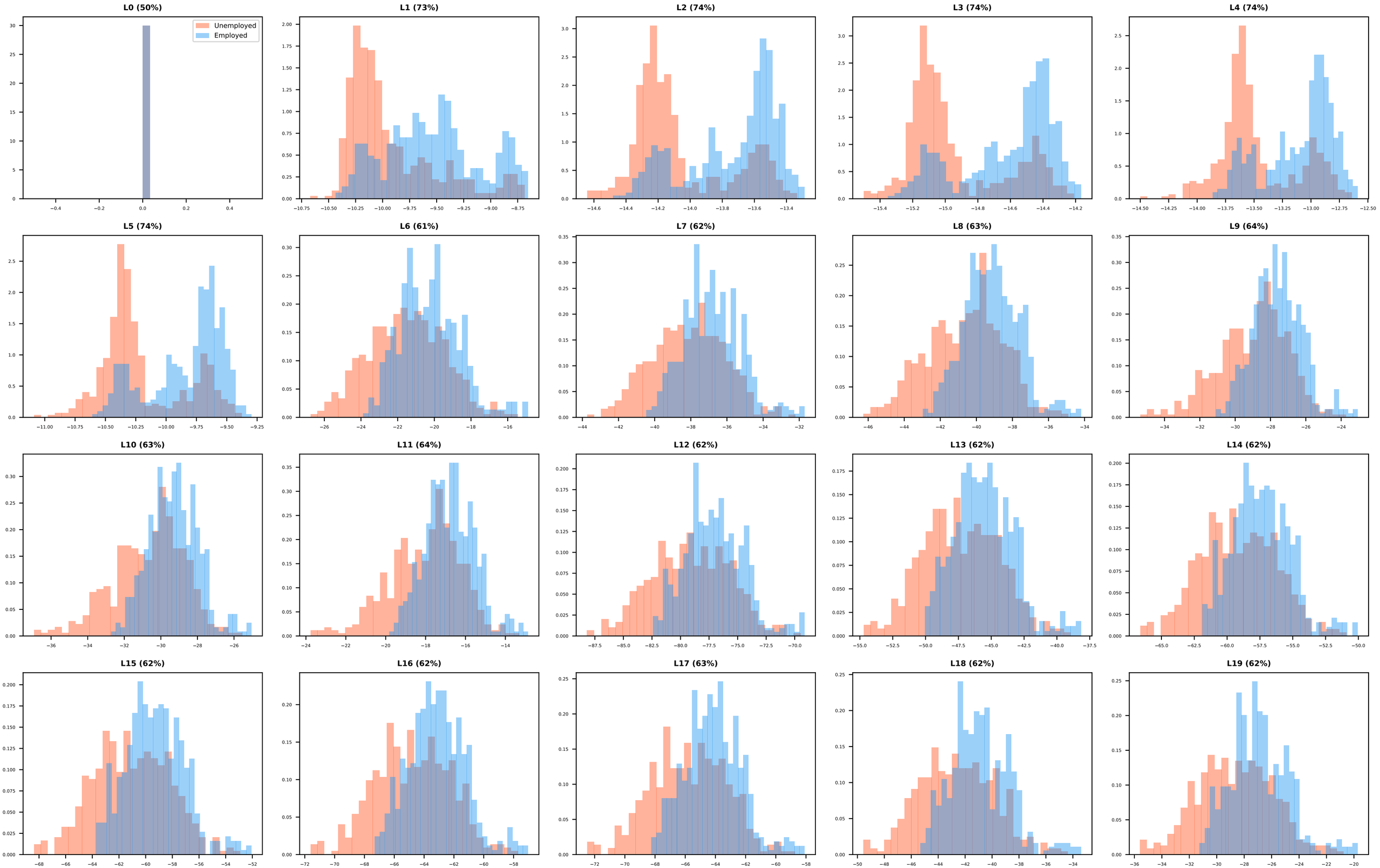
Projection Distribution — V8/exp2/cfg6 (geo=hash_day=orig_prompt=cot) — Page 3



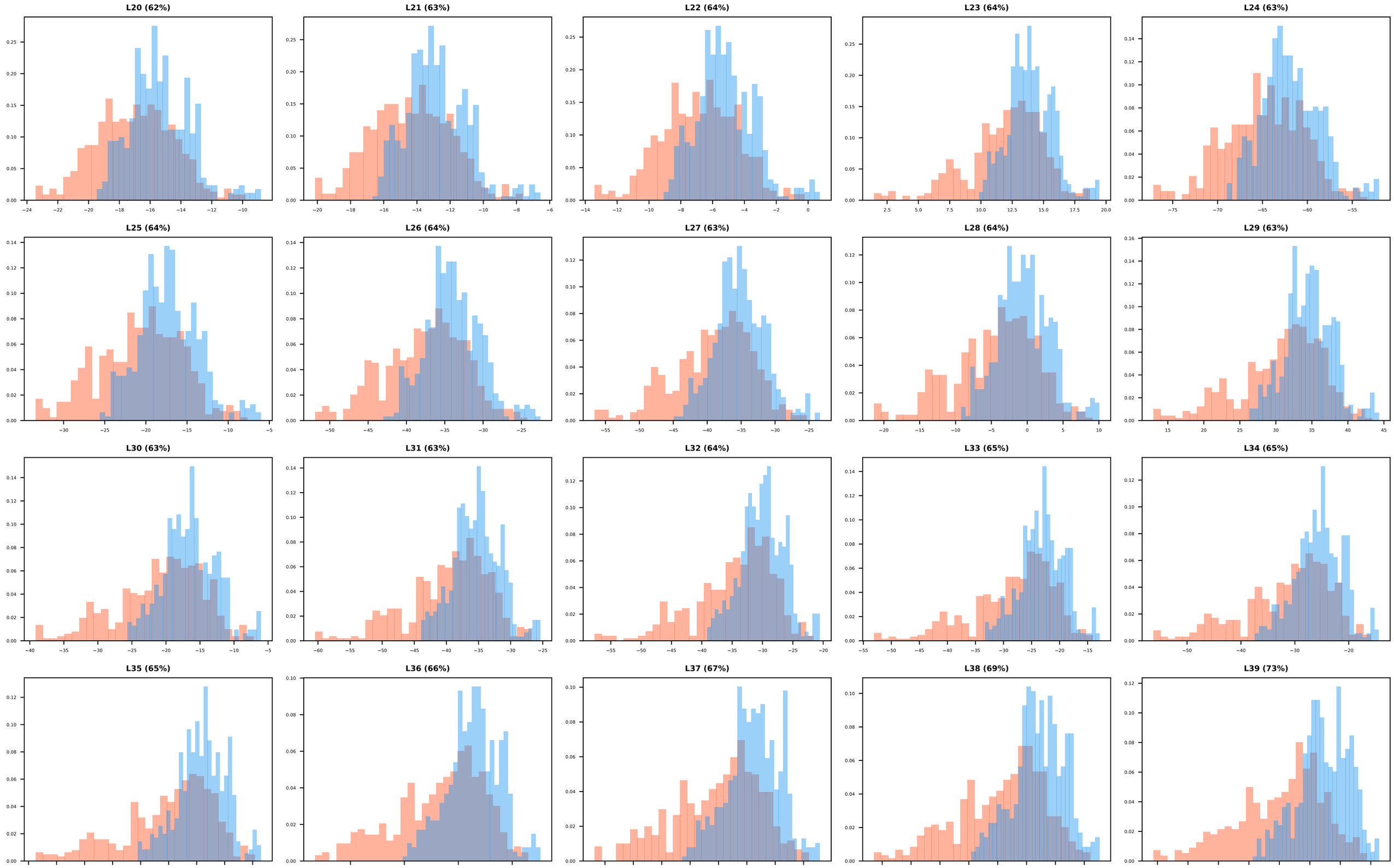
Projection Distribution — V8/exp2/cfg6 (geo=hash_day=orig_prompt=cot) — Page 4



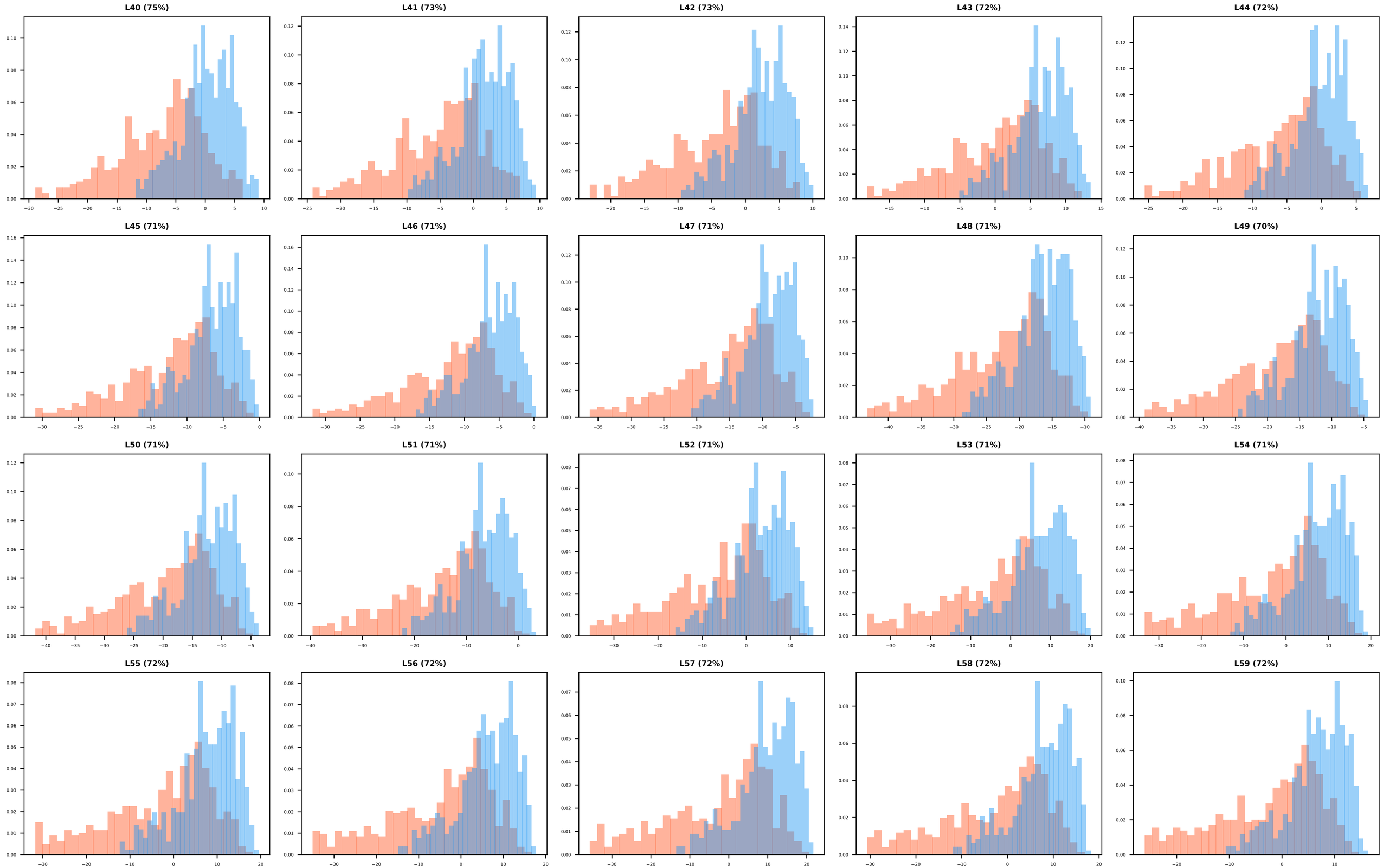
Projection Distribution — V8/exp2/cfg8 (geo=hash_day=hash_prompt=cot) — Page 1



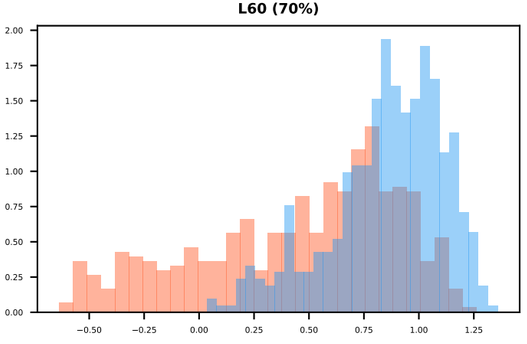
Projection Distribution — V8/exp2/cfg8 (geo=hash_day=hash_prompt=cot) — Page 2

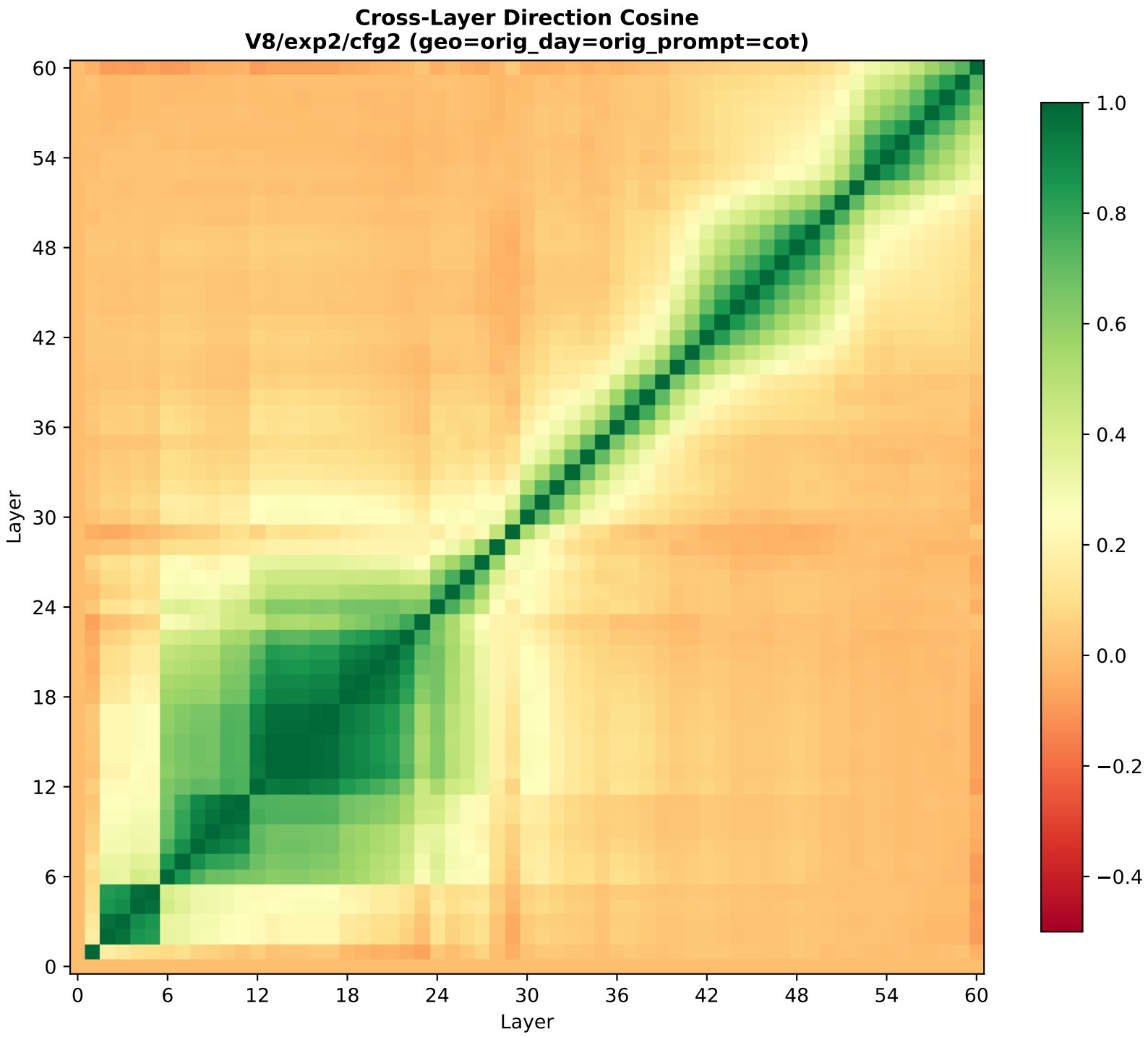


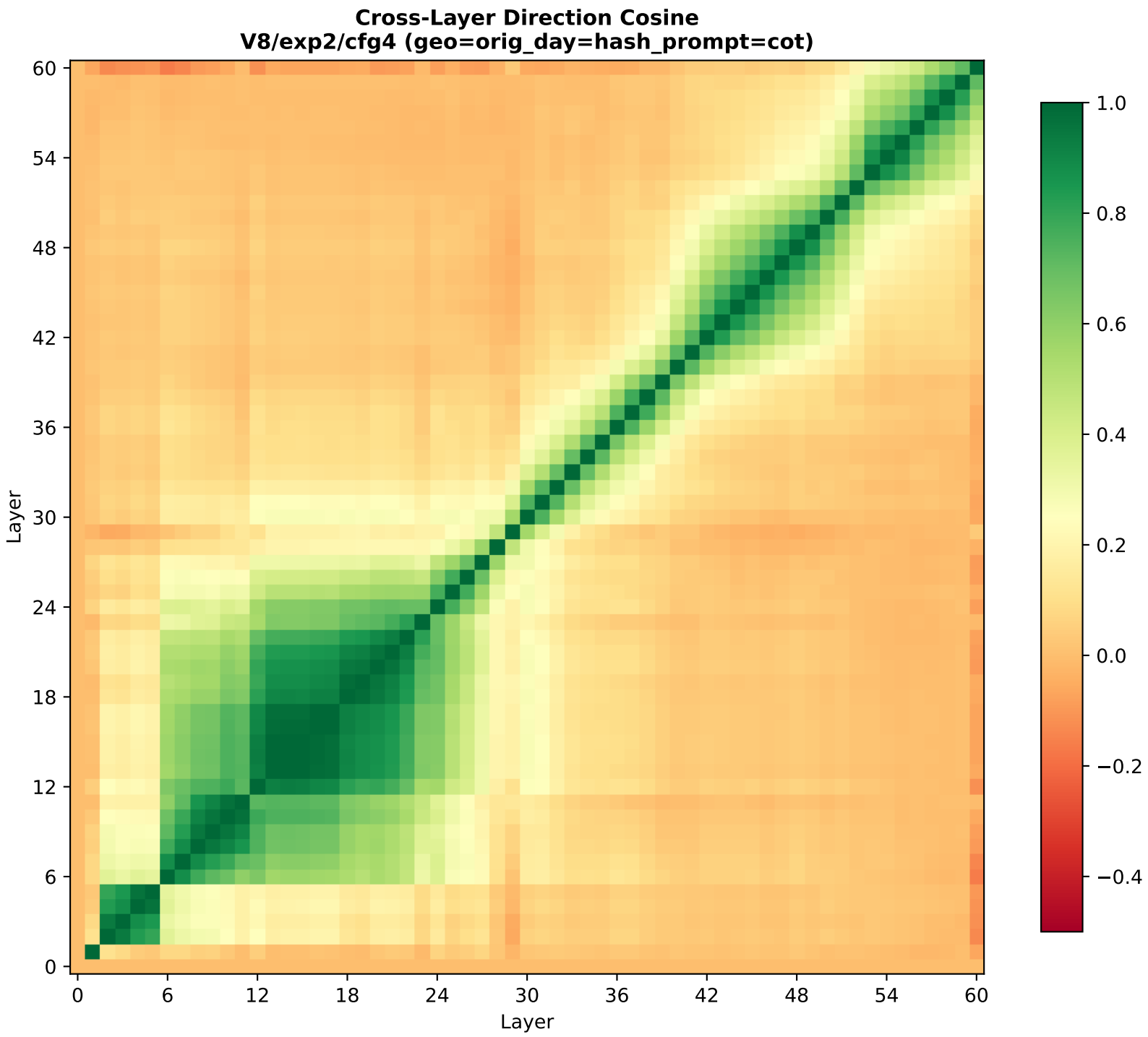
Projection Distribution — V8/exp2/cfg8 (geo=hash_day=hash_prompt=cot) — Page 3

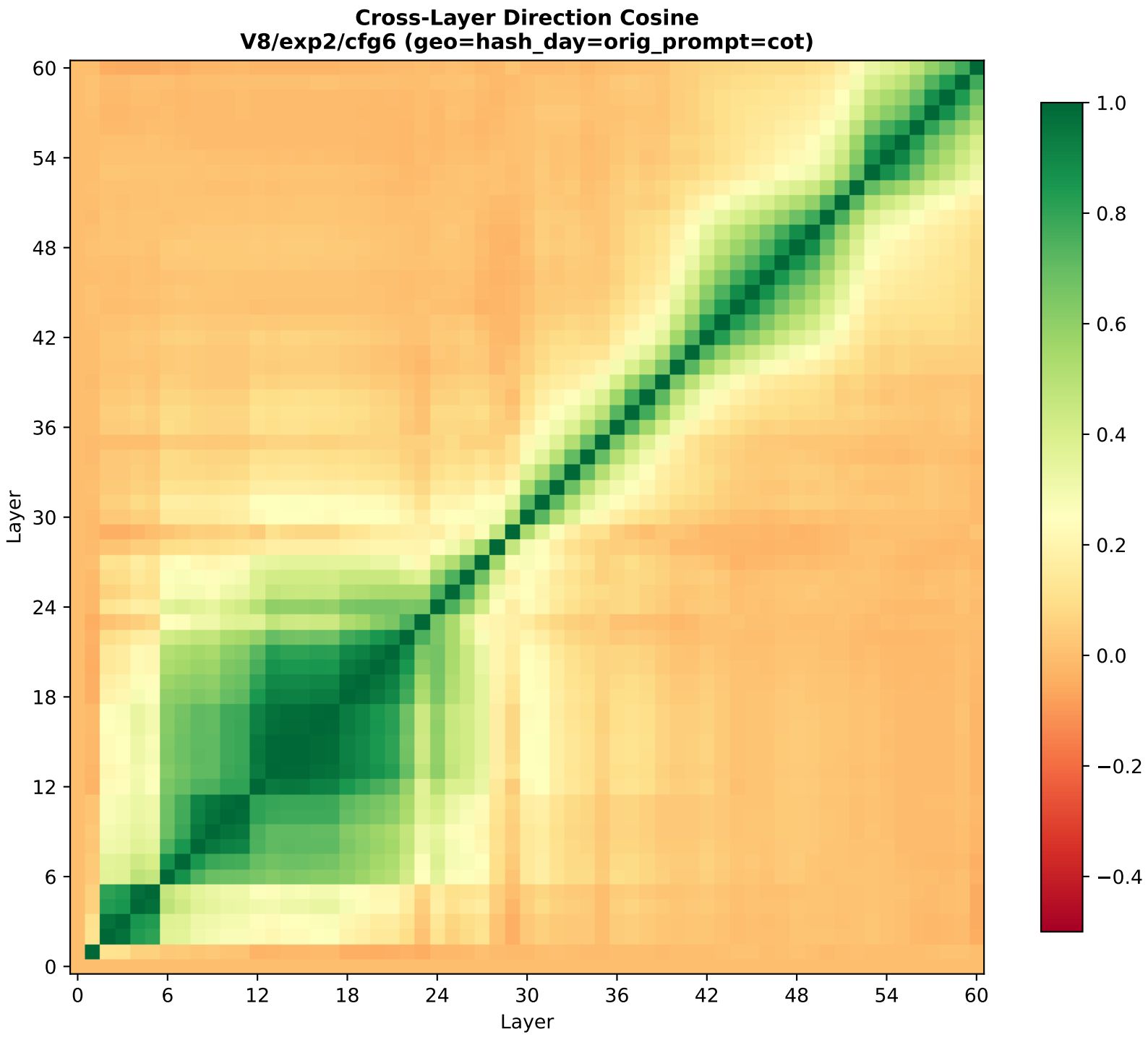


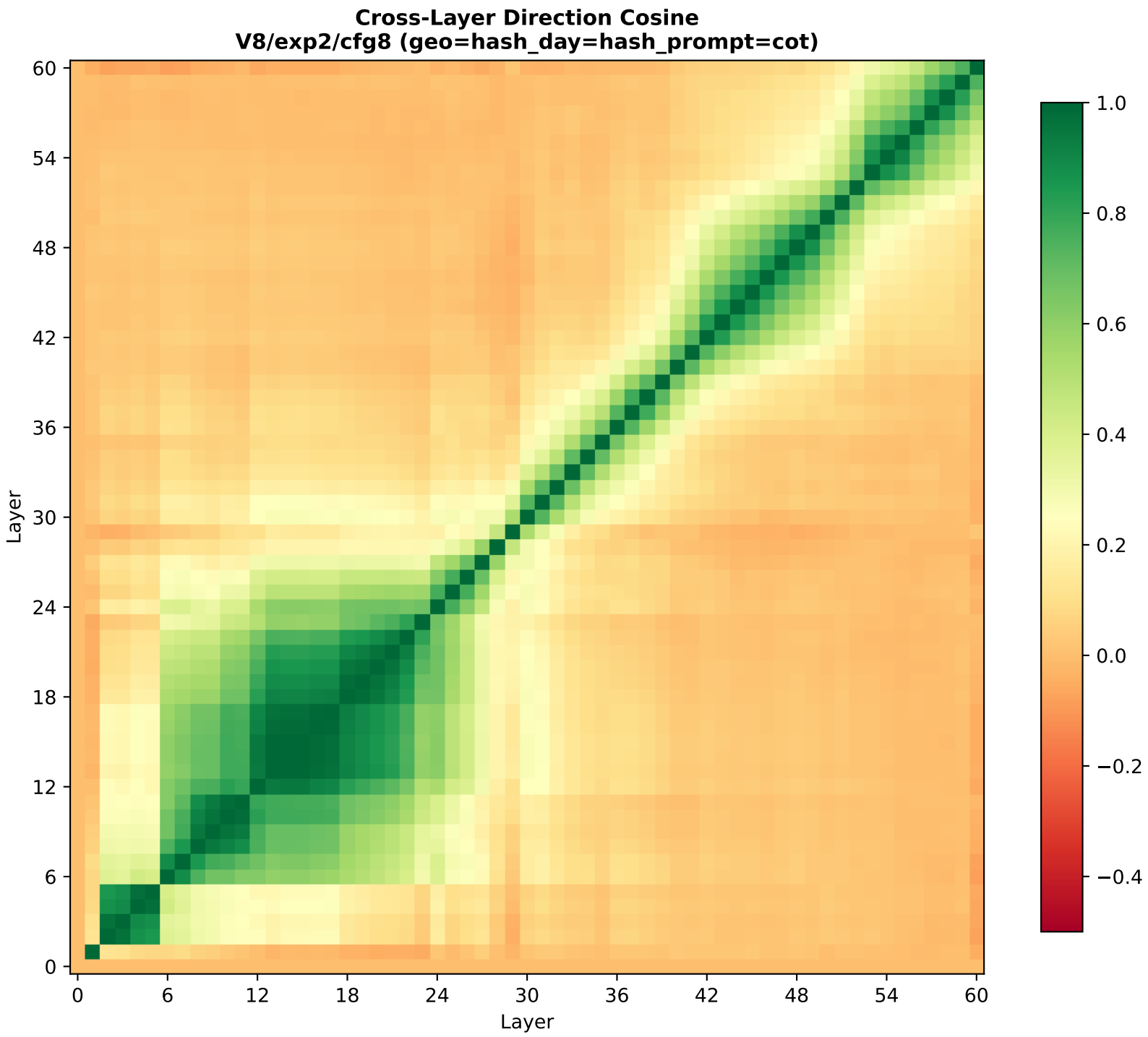
Projection Distribution — V8/exp2/cfg8 (geo=hash_day=hash_prompt=cot) — Page 4



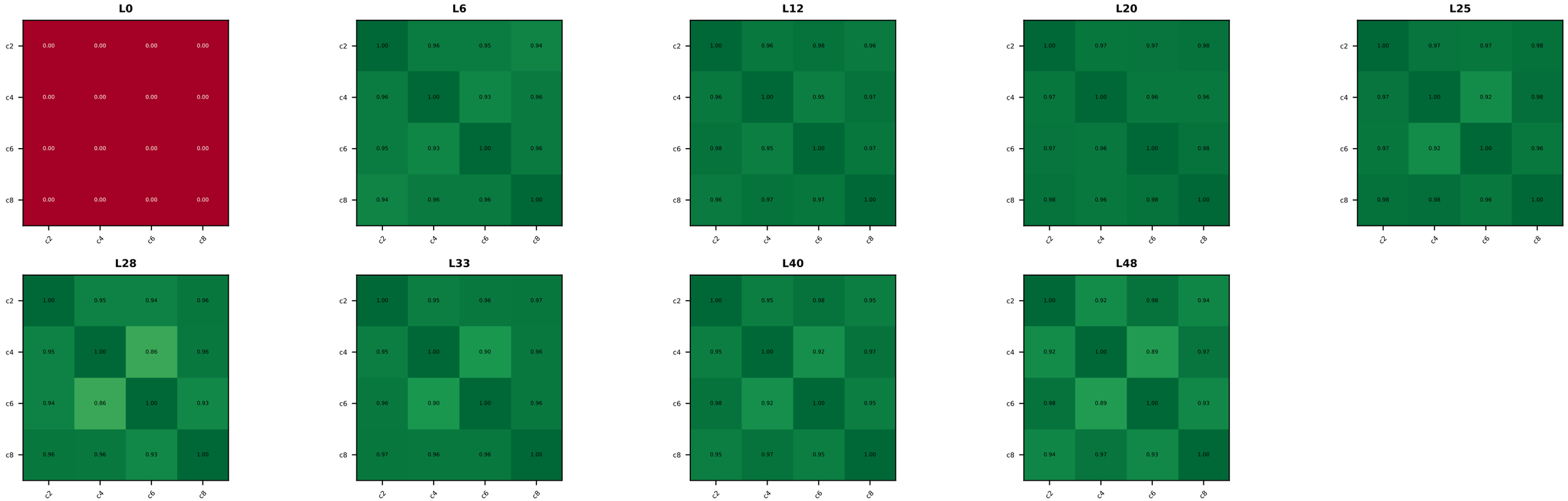

Cross-Layer Direction Cosine
V8/exp2/cfg2 (geo=orig_day=orig_prompt=cot)
Layer
Layer
0
6
12
18
24
30
36
42
48
54
60
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

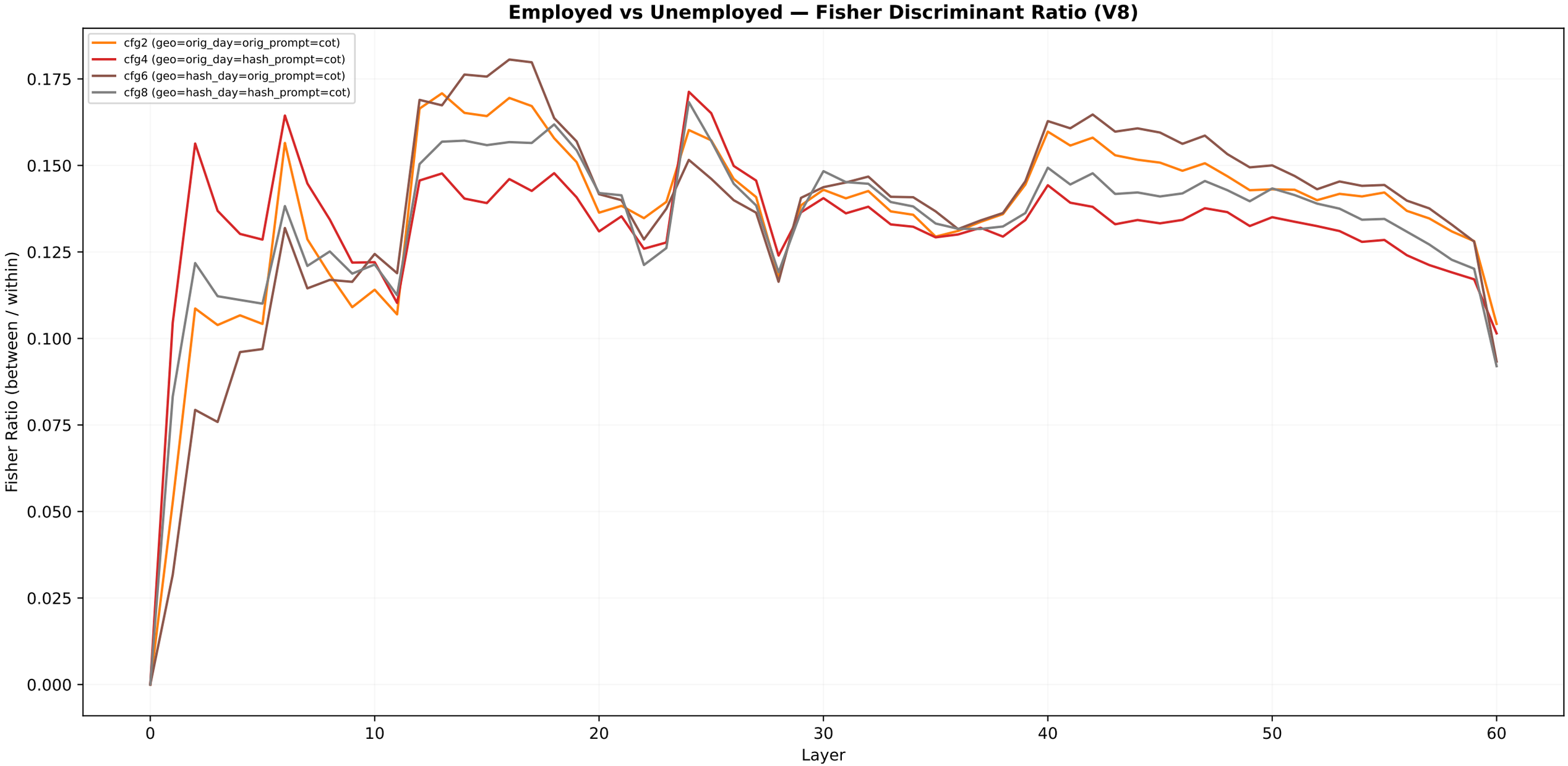

Cross-Layer Direction Cosine
V8/exp2/cfg4 (geo=orig_day=hash_prompt=cot)
Layer
Layer
0
6
12
18
24
30
36
42
48
54
60
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

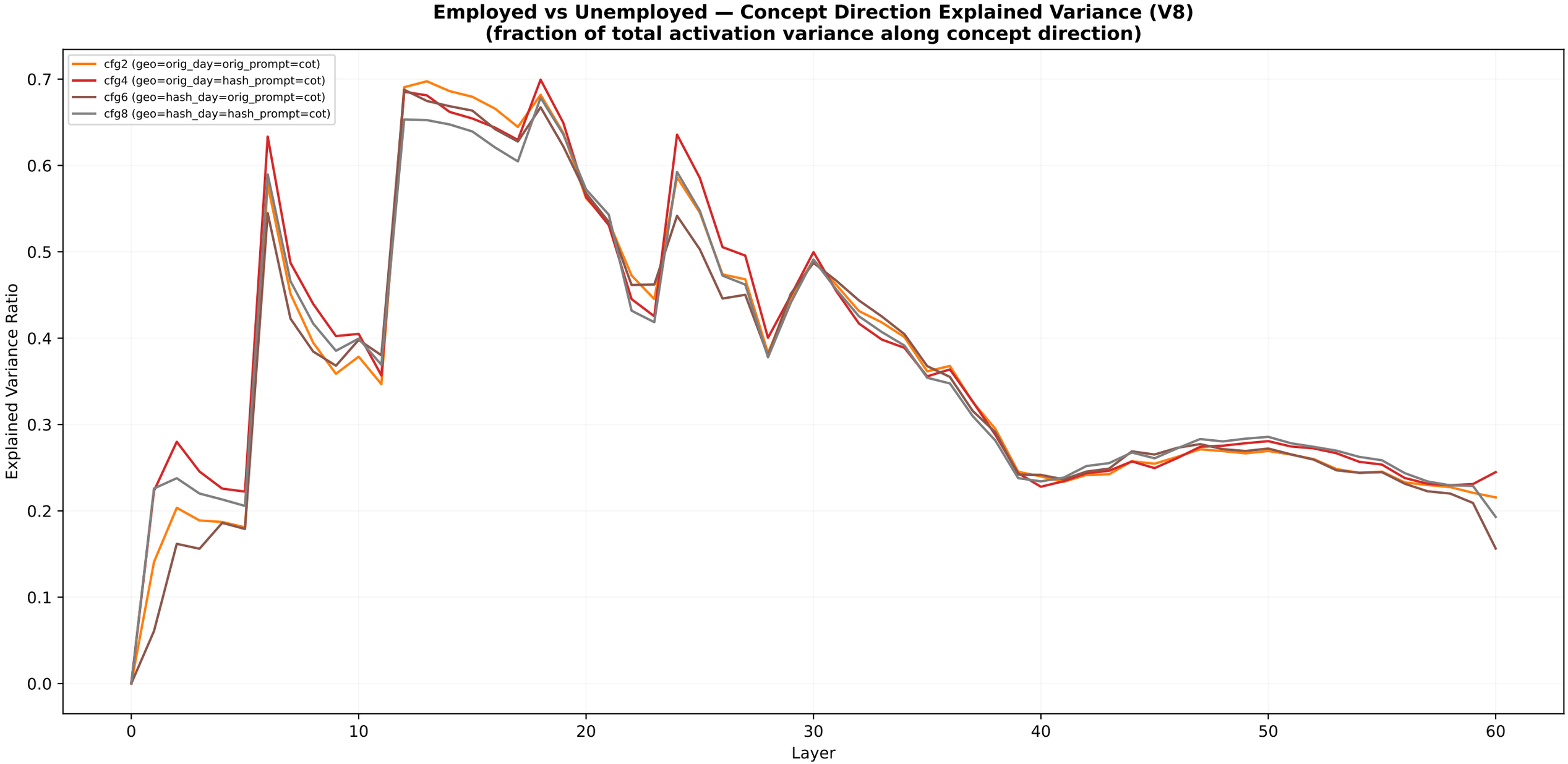

Cross-Layer Direction Cosine
V8/exp2/cfg6 (geo=hash_day=orig_prompt=cot)
Layer
Layer
0
6
12
18
24
30
36
42
48
54
60
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

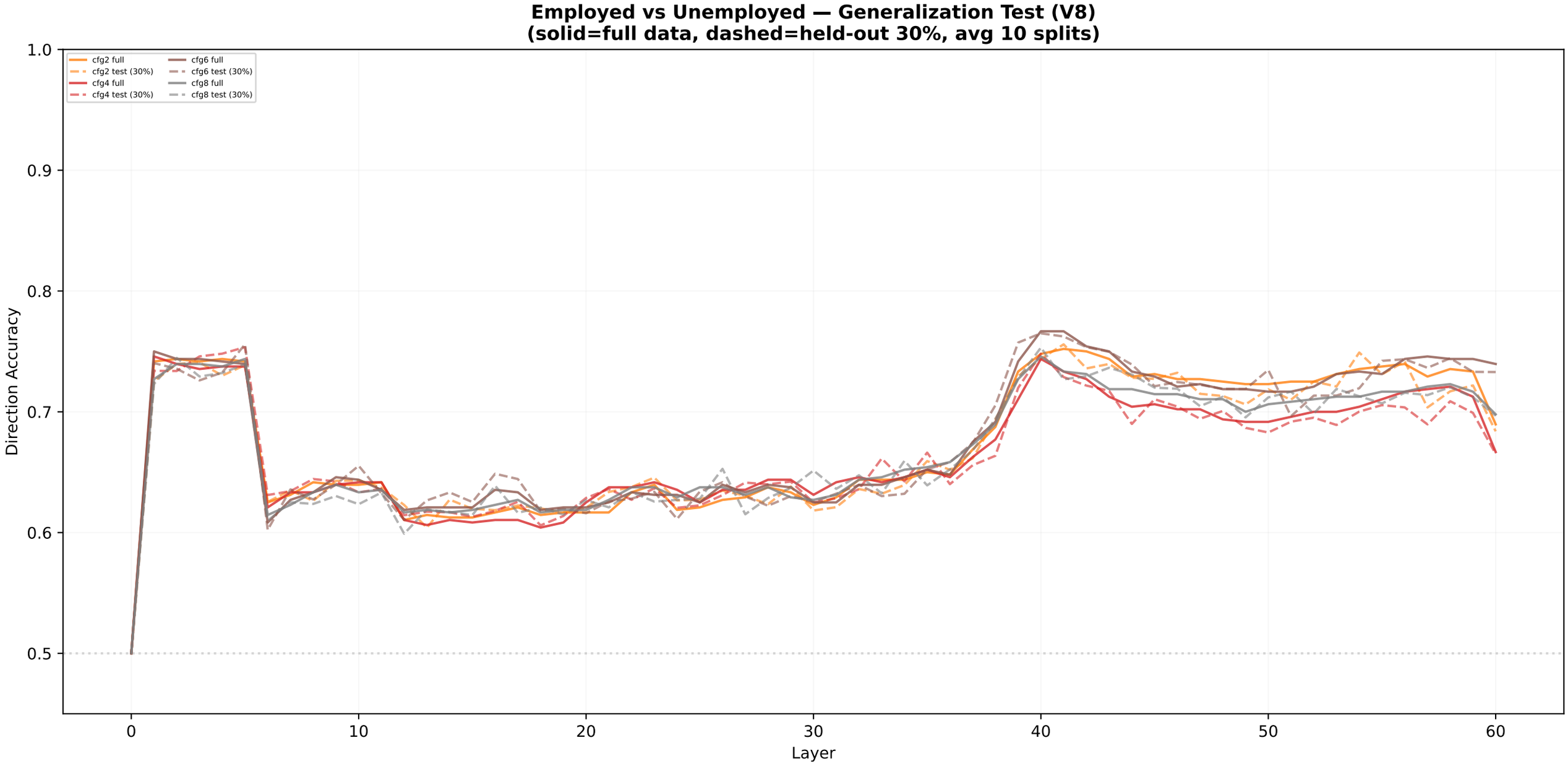

Cross-Layer Direction Cosine
V8/exp2/cfg8 (geo=hash_day=hash_prompt=cot)
Layer
Layer
0
6
12
18
24
30
36
42
48
54
60
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

Cross-Config Direction Cosine — V8/exp2
(do different configs find the same concept direction?)




Employed vs Unemployed — Fisher Discriminant Ratio (V8)
cfg2 (geo=orig_day=orig_prompt=cot)
cfg4 (geo=orig_day=hash_prompt=cot)
cfg6 (geo=hash_day=orig_prompt=cot)
cfg8 (geo=hash_day=hash_prompt=cot)
Fisher Ratio (between / within)
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0
10
20
30
40
50
60
Layer


Employed vs Unemployed — Concept Direction Explained Variance (V8)
(fraction of total activation variance along concept direction)
cfg2 (geo=orig_day=orig_prompt=cot)
cfg4 (geo=orig_day=hash_prompt=cot)
cfg6 (geo=hash_day=orig_prompt=cot)
cfg8 (geo=hash_day=hash_prompt=cot)
Explained Variance Ratio
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0
10
20
30
40
50
60
Layer


Employed vs Unemployed — Generalization Test (V8)
(solid=full data, dashed=held-out 30%, avg 10 splits)
cfg2 full
cfg2 test (30%)
cfg4 full
cfg4 test (30%)
cfg6 full
cfg6 test (30%)
cfg8 full
cfg8 test (30%)
Direction Accuracy
1.0
0.9
0.8
0.7
0.6
0.5
0
10
20
30
40
50
60
Layer

Geometry Summary — V8/exp2

Employed vs Unemployed — Geometry Summary (V8)

Cfg	Label	DA	@L	Sep	@L	Fisher	@L	EVR	@L	DA_gen	@L
2	geo=orig_day=orig_prompt=cot	75.2%	41	0.17411	57	0.17085	13	0.697492	13	76.1%	42
4	geo=orig_day=hash_prompt=cot	74.6%	1	0.16695	57	0.17129	24	0.699383	18	74.7%	2
6	geo=hash_day=orig_prompt=cot	76.7%	40	0.18159	57	0.18062	16	0.687762	12	76.8%	42
8	geo=hash_day=hash_prompt=cot	74.6%	40	0.17038	57	0.16832	24	0.678488	18	75.6%	5

DA: best direction accuracy (all data). Sep: max normalized centroid separation.
Fisher: max between/within variance ratio. EVR: max explained variance ratio.
DA_gen: best generalization DA (70/30 split, avg 5 trials).